

5-Day Roadmap: Stock Portfolio Performance Analysis

Using Python, SQL & Power BI — Condensed Sprint Plan

This is a compressed 5-day version of the full end-to-end Finance Data Analyst project. Each day groups multiple roadmap steps together, sized to be realistically achievable in a focused work day (roughly 6-8 hours). Heavier days (Day 2 and Day 4) are flagged.

Day	Focus	Original Steps Covered
Day 1	Plan the project + collect stock data	Step 1 - Step 2
Day 2	Clean the data + explore it (heavy day)	Step 3 - Step 4
Day 3	Build performance metrics + load into SQL	Step 5 - Step 6
Day 4	SQL analysis + Power BI dashboard (heavy day)	Step 7 - Step 8
Day 5	Write insights report + publish to GitHub	Step 9 - Step 10

DAY 1 — Define Project + Collect Stock Data

Step 1: Define Project Objective & Business Questions

Estimated time: ~1 hour

Set the scope before touching any data so every later step has a clear target.

Key questions to answer

- Which stocks performed best?
- What is the overall portfolio return?
- Which stocks carry the highest risk?
- Which sectors performed better?
- What investments contributed most to portfolio growth?

Output: A one-page project scope document + list of KPIs and business questions

Step 2: Collect Stock Market Data

Estimated time: ~2-3 hours

Use Python and the yfinance library to pull historical daily price data.

Tools

- Python
- Yahoo Finance API (yfinance)

Example tickers to collect

- Apple (AAPL), Microsoft (MSFT), NVIDIA (NVDA), Amazon (AMZN), Tesla (TSLA)
- RBC (RY.TO), TD Bank (TD.TO), Shopify (SHOP.TO)

Required fields

- Date, Stock Ticker, Open, High, Low, Close, Trading Volume

Output: Raw stock market dataset (CSV)

DAY 2 — Clean the Data + Explore It (heaviest day)

Step 3: Data Cleaning Using Python

Estimated time: ~2 hours

Cleaning tasks

- Check for missing values
- Remove duplicates
- Convert date formats
- Fix incorrect data types
- Check for outliers
- Standardize stock symbols

Python libraries

- Pandas
- NumPy

Output: Clean stock dataset (CSV)

Step 4: Exploratory Data Analysis (EDA)

Estimated time: ~3-4 hours

The most time-intensive step of the sprint — budget extra time here.

Price analysis

- Stock price trends
- Highest and lowest prices
- Growth patterns

Return analysis

- Daily returns
- Monthly returns
- Annual returns

Risk analysis

- Volatility
- Standard deviation
- Maximum drawdown

Relationship analysis

- Correlation between stocks
- Sector comparison

Python libraries

- Matplotlib
- Seaborn

Output: Analysis charts + financial metrics dataset

DAY 3 — Build Metrics + Load Into SQL

Step 5: Create Financial Performance Metrics

Estimated time: ~1.5 hours

Portfolio metrics

- Initial investment
- Current portfolio value
- Total return %
- Profit/Loss

Stock metrics

- Return %
- Volatility
- Sharpe Ratio
- Risk level
- Performance ranking

Output: Stock performance summary table

Step 6: Store Data in a SQL Database

Estimated time: ~2-3 hours

SQLite is the fastest option for a solo portfolio project — no server setup required.

Suggested tables

- Stock_Prices — Date, Ticker, Open, High, Low, Close, Volume
- Company_Master — Ticker, Company Name, Sector, Industry
- Portfolio — Stock, Number of Shares, Purchase Price, Investment Amount

Output: Structured financial database

DAY 4 — SQL Analysis + Power BI Dashboard (heaviest day)

Step 7: Perform SQL Analysis

Estimated time: ~2 hours

Stock performance queries

- Top-performing stocks
- Lowest-performing stocks
- Average returns

Portfolio analysis queries

- Total portfolio value
- Profit/loss by stock
- Contribution of each stock

Market analysis queries

- Monthly performance
- Sector comparison
- Trading volume trends

SQL techniques to use

- SELECT
- JOIN
- GROUP BY
- Window Functions
- Aggregate Functions

Output: SQL analysis results

Step 8: Build the Power BI Dashboard

Estimated time: ~4-5 hours

Connect Power BI to your SQL database (or import CSVs) and build four pages.

Page 1: Portfolio Overview

- KPIs: Total Investment, Current Value, Total Return %, Profit/Loss, Best Performing Stock
- Visuals: Portfolio growth trend, performance ranking

Page 2: Stock Performance Analysis

- Stock return comparison, price movement charts, top 5 performing stocks, monthly performance

Page 3: Risk Analysis

- Risk vs return scatter plot, volatility comparison, Sharpe Ratio ranking, max drawdown analysis

Page 4: Sector Analysis

- Technology vs Banking vs Consumer comparison, sector returns, trading volume analysis

Output: Power BI dashboard file (.pbix)

DAY 5 — Insights Report + Publish the Project

Step 9: Create a Business Insights Report

Estimated time: ~2 hours

Document findings on

- Which stocks generated the highest returns?
- Which stocks had the highest risk?
- Which sector performed best?
- What investment strategy could improve returns?

Include

- Key findings
- Charts
- Recommendations

Output: Business insights report (PDF or Word doc)

Step 10: Publish the Project Portfolio

Estimated time: ~1-2 hours

Package everything into a clean, professional GitHub repository.

Suggested repo structure

- Data/ → Raw_Data, Clean_Data
- Python/ → Data_Collection, Data_Cleaning, Analysis
- SQL/ → Queries
- PowerBI/ → Dashboard
- README.md

Output: A public GitHub repository link ready to share

Reference Links

Category	Resource	Link
Stock Data API	Yahoo Finance	finance.yahoo.com
Python Library Docs	yfinance	pypi.org/project/yfinance
Python Library Docs	Pandas	pandas.pydata.org/docs
Python Library Docs	NumPy	numpy.org/doc
SQL Practice	SQLBolt	sqlbolt.com
SQL Practice	PostgreSQL Docs	postgresql.org/docs
Power BI Learning	Microsoft Power BI Learning	learn.microsoft.com/power-bi
Financial Concepts	Investopedia	investopedia.com

Final Project Stack

Stage	Tool
Data Collection	Python + yfinance
Data Cleaning	Python (Pandas)
Analysis	Python
Database	SQL
Visualization	Power BI
Documentation	GitHub

This 5-day sprint compresses the full Finance Data Analyst roadmap into a focused, achievable schedule — ideal for a portfolio project built over one intensive work week.